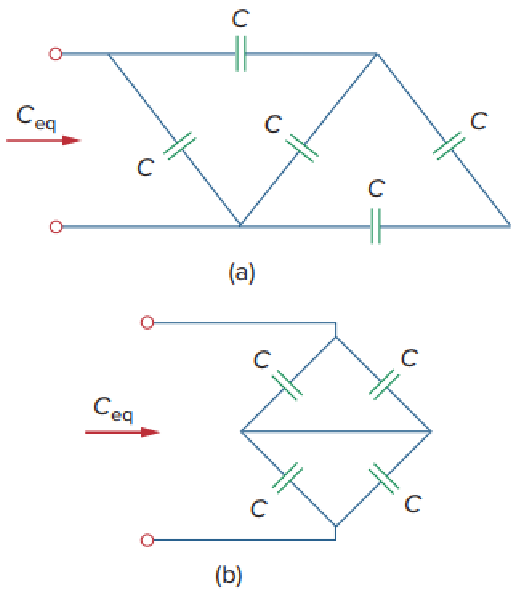
Chapter 6, Problem 29P

Problem

Determine *Ceq* for each circuit in Fig. 6.61.

Fig. 6.61:



Step-by-step solution

Step 1 of 4

(a)

Refer to Figure 6.61 (a) in the textbook.

In Figure 6.61 (a), the series combination of two capacitors connected in parallel with a capacitor, C. Calculate their equivalent capacitance.

$$\begin{aligned} C_{eq1} &= \left(\frac{C\times C}{C+C}\right)+C \\ &= \frac{C}{2}+C \\ &= \frac{3C}{2} \end{aligned}$$

Draw the reduced circuit of Figure 6.61(a).

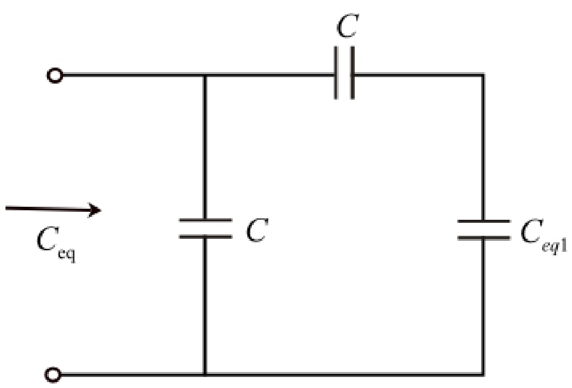


Figure 1

Step 2 of 4

In Figure 1, the capacitors C_{eq1} and C are connected in series and their resultant is in parallel with capacitor, C.

Calculate the equivalent capacitance.

$$\begin{aligned}C_{\text{eq}} &= \frac{C_{eq1}\times C}{C_{eq1}+C}+C\\ &= \frac{\frac{3}{2}C\times C}{\frac{3}{2}C+C}+C\\ &= \frac{\frac{3}{2}C}{\frac{5}{2}}+C\\ &= \frac{3}{5}C+C\end{aligned}$$

$$=\frac{8}{5}C$$
$$=1.6C$$

Therefore, the equivalent capacitance, C_{eq} is $\boxed{1.6C}$.

Step 3 of 4

(b)

Refer to Figure 6.61 (b) in the textbook.

In Figure 6.61 (b), the capacitors are connected in parallel. Calculate their equivalent capacitances.

$$\begin{aligned}C_{eq1} &= C+C\\ &= 2C\\ C_{eq2} &= C+C\\ &= 2C\end{aligned}$$

Draw the reduced circuit of Figure 6.61(b).

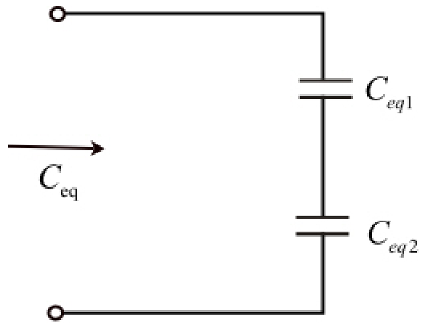


Figure 2

Step 4 of 4

Calculate the equivalent capacitance.

$$\begin{aligned} C_{\text{eq}} &= \frac{(C_{eq1})\times(C_{eq2})}{C_{eq1}+C_{eq2}} \\ &= \frac{(2C)\times(2C)}{2C+2C} \\ &= \frac{4C^2}{4C} \\ &= C \end{aligned}$$

Therefore, the equivalent capacitance, C_{eq} is $\boxed{C}$.